# HIL-based Augmented ICS (HAI) Security Dataset v1.0

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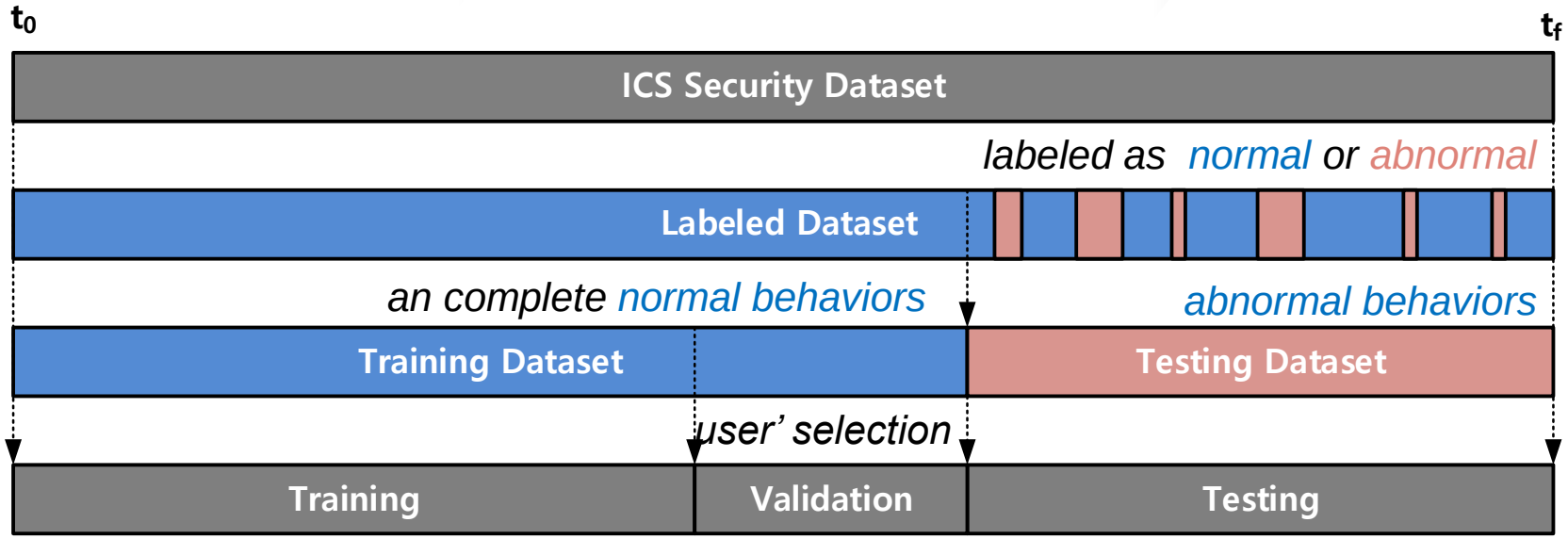
# Introduction

## An Industrial Control System (ICS) Security Dataset

- Essential to develop ICS security research based on AI techniques
- A labeled time series data that is collected on both normal & abnormal situations of ICS

### General Scheme for AI-based security research

| Training Stage | Validation Stage | Testing Stage |
|---|---|---|
| • Extraction of the ICS features<br>• Training to fit a model using training data | • Tuning the hyper parameters<br>• Selection of the best model | • Prediction and evaluation of the model using various metric |



# Introduction

*HAI 1.0 focused on*




- **Overcoming the process simplicity of lab-scale testbeds**
- **Minimization of long-term human intervention for normal operations**

① **Process augmentation with a HIL simulator**

② **Unmanned normal Operation**



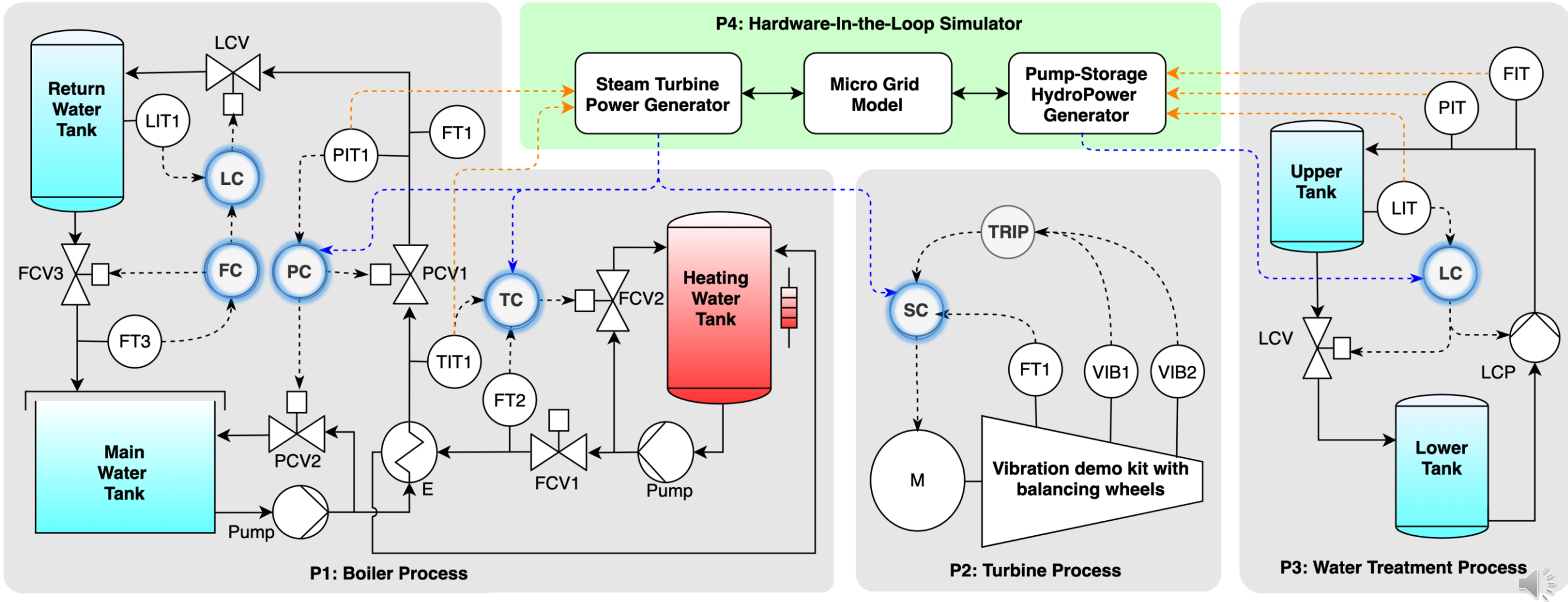

- **Realization of various & sophisticated ICS attacks on real-world system**
  - Labeling anomalies accurately
  - Maintaining consistency for replicates
  - Being able to systematically expand the attacks on a large-scale system

③ **Scalable attack tool based on process control loop**

# HAI Testbed

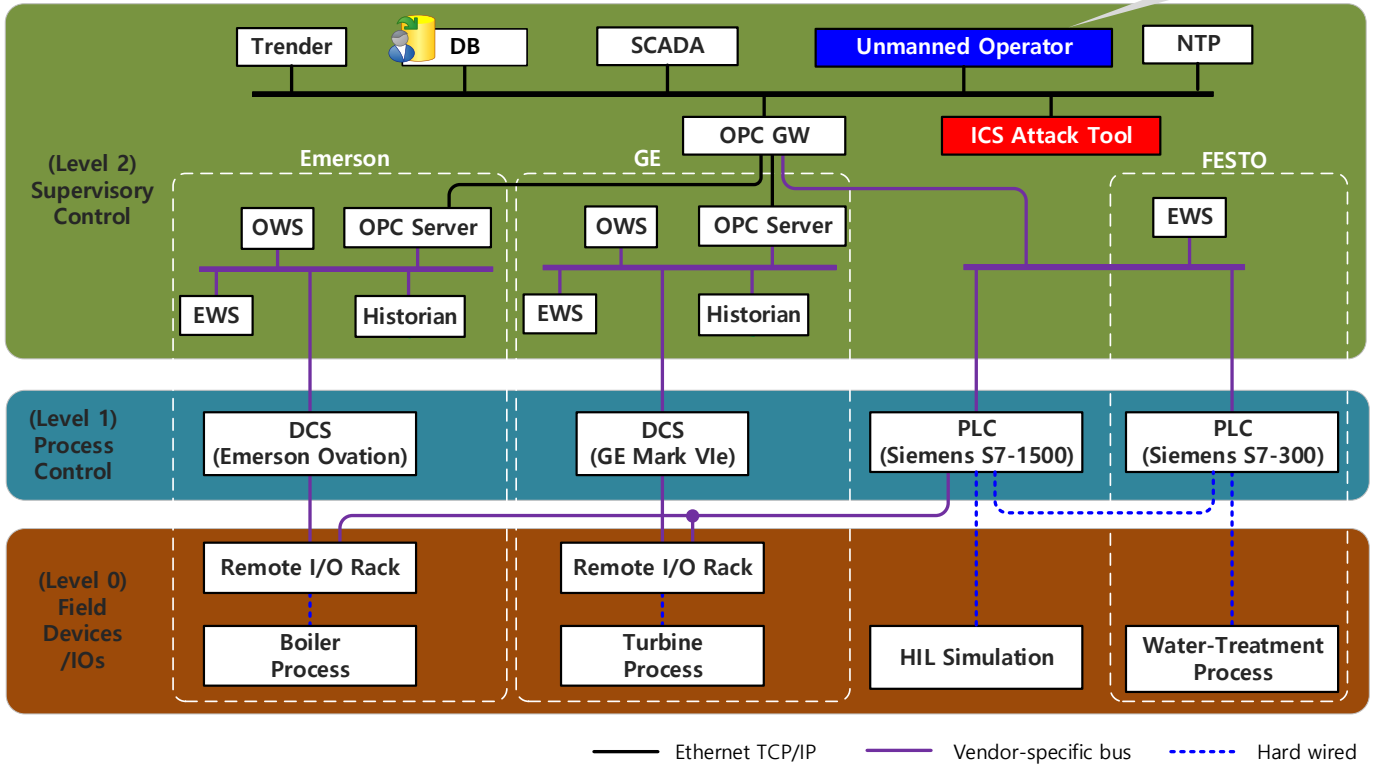
## ① ICS process augmentation with a HIL(Hardware-In-the-Loop) simulator

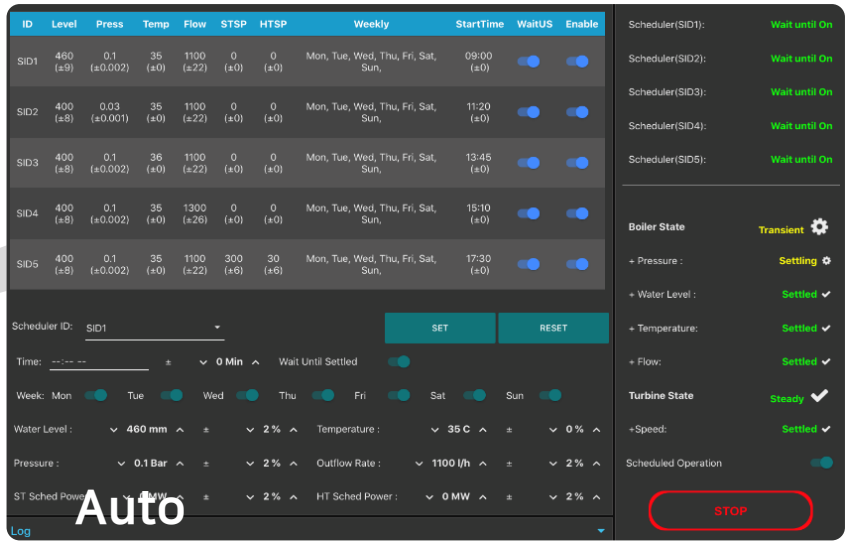
- Three ICS testbeds were interconnected via HIL simulator that simulates complex power generation system.
- To increase the correlation between signals, not to get precise simulation results

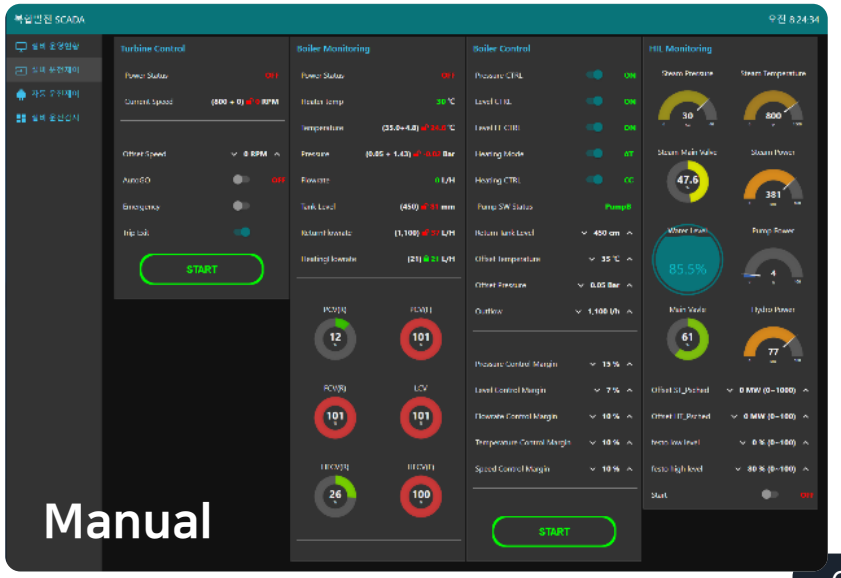


## ② Unmanned Normal Operation

- Changing the set points for five controllers (PC, LC, FC, TC, LC)
  - 5 times a day,  start with a random delay
- Automatic operation
  1) Check whether the controller is stabilized at the scheduled time
  2) Send a new SP command within operational range

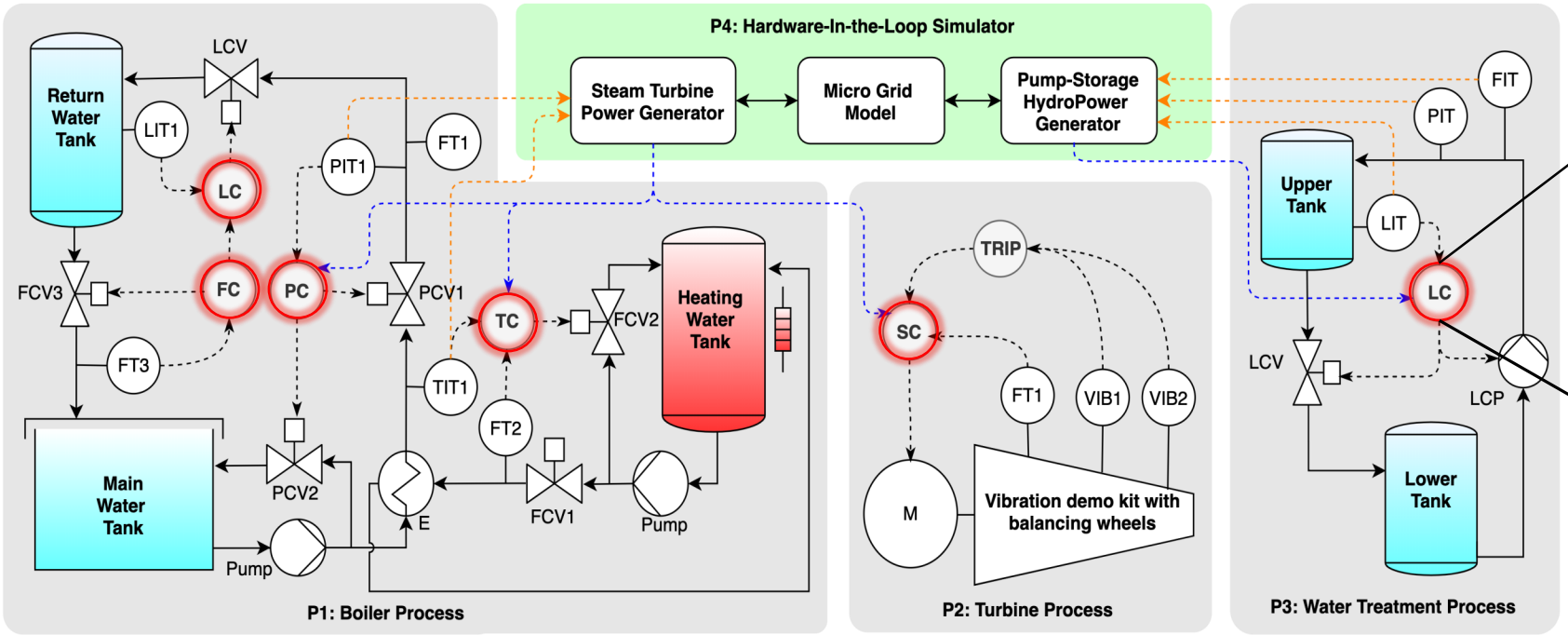


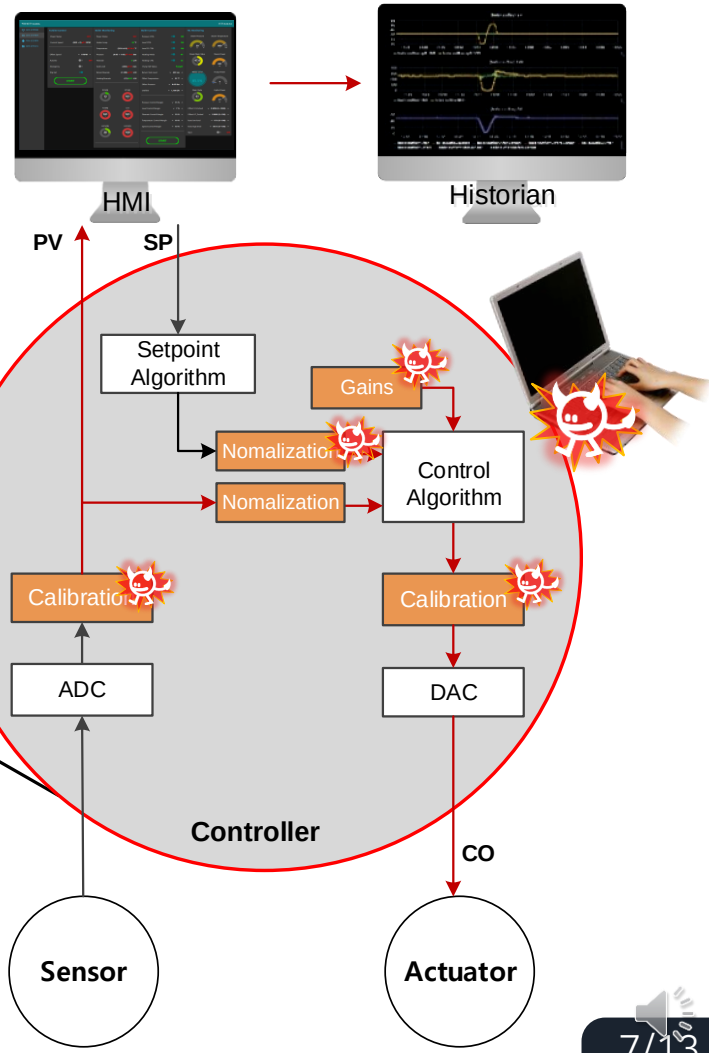



## ③ Scalable Attack Tool based on Process Control Loop (CSET '19)

- Attack targets: PCLs = {'LC', 'FC', 'PC', 'SC', 'LC'} x Variables:{'SP', 'PC', 'CO'}
- Changing the SP, PV, CO values by modifying the parameters of Function Block(FB)
  - Calibration FB: $y = ax + b$
  - Normalization FB: $y = \frac{x-a}{b-a}$
  - PID control algorithm FB: $y = Pe(t) + I\int e(t)dt + D\frac{de(t)}{dt},\ e(t) = PV(t) - SP(t)$



# HAI 1.0 Security Dataset

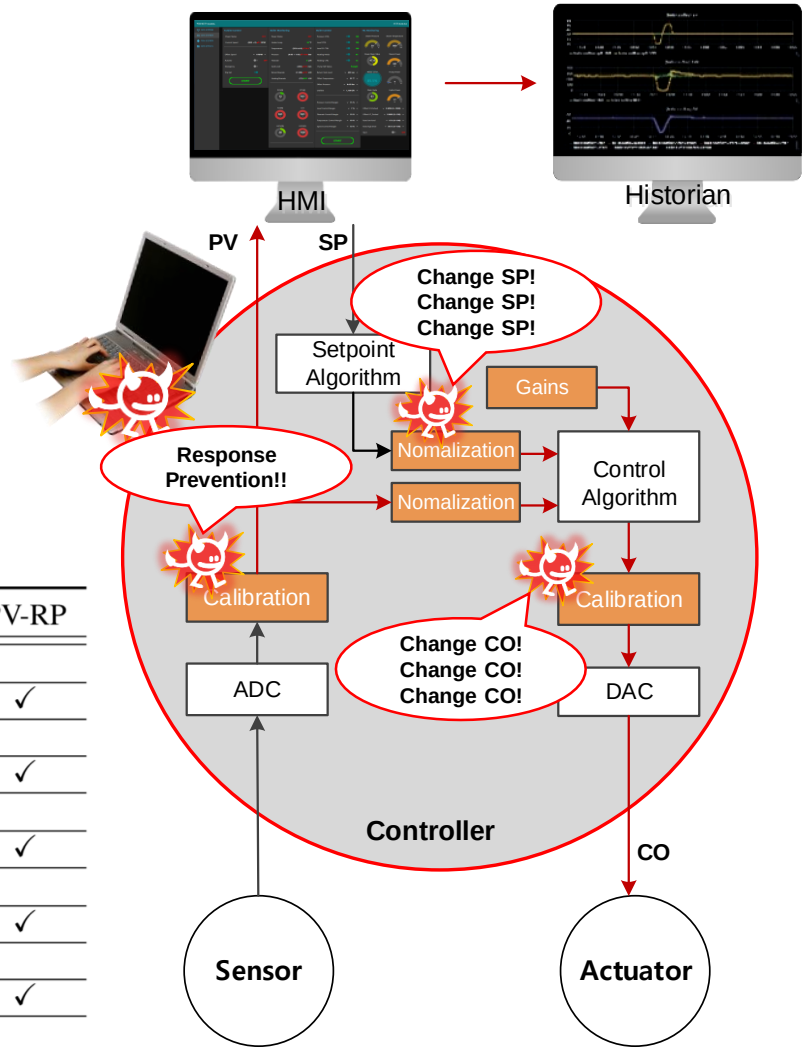
## ④ Attack Primitives Based on PCL

- Attack instances for a single PCL
- Attack scenario = combination of PCL attack primitives
- Attack types
  1) Response Prevention: hiding abnormal response on PV on HMI
  2) SP attack: forcing the SP value to indirectly change the CO value
  3) CO attack: forcing the CO value directly

## 14 Attack Primitives for HAI Testbed

- For five PCLs (P1.PC, P1.FC, P1.LC, P2. SC, P3.LC)
  - 4 SP attacks [1,5,7,11]
  - 4 SP&RP attacks [2,6,8,12]
  - 2 CO attacks [3,8]
  - 2 CO&RP attacks [4, 10]
  - 2 SP&CO attacks [13,14]

| No | Attack ID | Target | SP | CO | PV-RP |
|---|---|---|---|---|---|
| 1 | AP-P1PC-SP | P1.PC | ✓ | | |
| 2 | AP-P1PC-SPRP | | ✓ | | ✓ |
| 3 | AP-P1PC-CO | | | ✓ | |
| 4 | AP-P1PC-CORP | | | ✓ | ✓ |
| 5 | AP-P1FC-SP | P1.FC | ✓ | | |
| 6 | AP-P1FC-SPRP | | ✓ | | ✓ |
| 7 | AP-P1LC-SP | P1.LC | ✓ | | |
| 8 | AP-P1LC-SPRP | | ✓ | | ✓ |
| 9 | AP-P1LC-CO | | | ✓ | |
| 10 | AP-P1LC-CORP | | | ✓ | ✓ |
| 11 | AP-P2SC-SP | P2.SC | ✓ | | |
| 12 | AP-P2SC-SPRP | | ✓ | | ✓ |
| 13 | AP-P3LC-SP1CO1 | P3.LC | $✓_{LH}$ | $✓_{LCV}$ | |
| 14 | AP-P3LC-SP2CO2 | | $✓_{LL}$ | $✓_{LCP}$ | |



# HAI 1.0 Security Dataset

## Implementing 'AP-P1FC-SPRP' Attack Primitives

### 1. PCL Configuration

- PCL variables {SP='B3005', PV='FT01', CO='FCV01'}
- FB parameters of the PCL variables

### 2. Attack Configuration

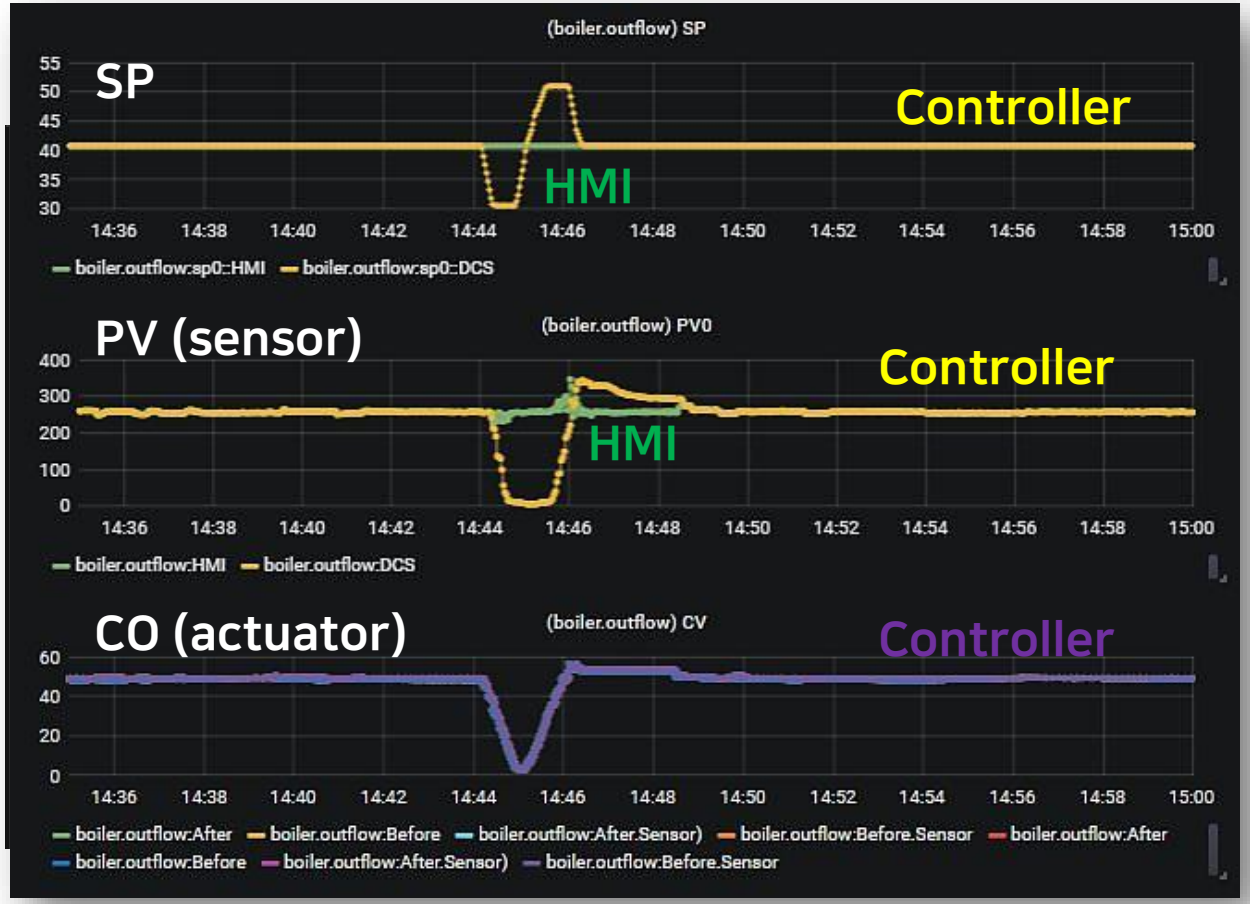
- Response prevention : replaying PV with a normal snapshot
- SP attack: manipulating the SP value hiding SP changes

### 3. Attack Scheduling

- Attack task starts at the scheduled time

### 4. Data Labeling

- Detecting the forced changes of FB parameters
- Extracting the attack interval and points
  (e.g. 'Boiler-FC- SP', 'Boiler-FC-PV')



# HAI 1.0 Security Dataset

## Two Dataset

- **Dataset A**
  - Training: 7 day
  - Testing: 28 attacks over 4 days

- **Dataset B**
  - Training: 3 days
  - Testing: 10 attacks over 1.5 days

**Training dataset (7 days)**

| No | Attack IDs | Start Time | Duration |
|---|---|---|---|
| 1 | AP-P1LC-SPRP | 19-10-29 13:40 | 370 secs |
| 2 | AP-P1LC-CO | 19-10-29 14:35 | 312 secs |
| 3 | AP-P1LC-CORP | 19-10-29 15:45 | 868 secs |
| 4 | AP-P1FC-CORP | 19-10-29 16:30 | 262 secs |
| 5 | AP-P1LC-SP | 19-10-30 08:50 | 371 secs |
| 6 | AP-P1PC-SP | 19-10-30 09:40 | 334 secs |
| 7 | AP-P1PC-SPRP | 19-10-30 10:35 | 504 secs |
| 8 | AP-P1PC-CO | 19-10-30 11:37 | 268 secs |
| 9 | AP-P1PC-CORP | 19-10-30 12:30 | 518 secs |
| 10 | AP-P2SC-SPRP | 19-10-30 14:30 | 370 secs |
| 11 | AP-P3LC-SP2CO2 | 19-10-30 15:35 | 180 secs |
| 12 | AP-P3LC-SP1CO1 | 19-10-30 16:33 | 154 secs |
| 13 | AP-P2SC-SP | 19-10-31 08:42 | 348 secs |
| 14 | AP-(P1PC-SPRP, P2SC-SPRP) | 19-10-31 10:30 | 518 secs |
| 15 | AP-(P1PC-CO, P2SC-SP) | 19-10-31 11:33 | 346 secs |
| 16 | AP-P2SC-SPRP [Repeat No.11] | 19-10-31 13:25 | 368 secs |
| 17 | AP-(P1LC-CORP, P2SC-SPRP) | 19-10-31 14:30 | 396 secs |
| 18 | AP-(P1FC-SP, P2SC-SP) | 19-10-31 15:41 | 348 secs |
| 19 | AP-(P1PC-SP, P3LC-SP1CO1) | 19-10-31 16:30 | 398 secs |
| 20 | AP-(P1LC-SPRP, P3LC-SP1CO1) | 19-11-01 09:29 | 560 secs |
| 21 | AP-(P1LC-CO, P3LC-SP1CO1) | 19-11-01 10:41 | 310 secs |
| 22 | AP-P3LC-SP1CO1 [Repeat No.12] | 19-11-01 11:23 | 180 secs |
| 23 | AP-(P1FC-SPRP, P1LC-SP) | 19-11-01 12:31 | 506 secs |
| 24 | AP-(P1PC-CO, P1FC-SPRP) | 19-11-01 13:41 | 580 secs |
| 25 | AP-P1PC-SP [Repeat No.6] | 19-11-01 14:23 | 310 secs |
| 26 | AP-(P1FC-SP, P1PC-SPRP) | 19-11-01 15:31 | 520 secs |
| 27 | AP-P1FC-SPRP | 19-11-01 16:18 | 560 secs |
| 28 | AP-(P1PC-SPRP, P3LC-SP2CO2) | 19-11-01 17:20 | 520 secs |

**Training dataset (3 days)**

| | | | |
|---|---|---|---|
| 29 | AP-(P1FC-SP, P1PC-SP) | 19-11-04 15:31 | 410 secs |
| 30 | AP-(P1PC-SP, P3LC-SP2CO2) | 19-11-04 17:20 | 520 secs |
| 31 | AP-(P1LC-CO, P3LC-SP2CO2) | 19-11-05 09:30 | 380 secs |
| 32 | AP-(P1FC-SP, P3LC-SP2CO2) | 19-11-05 10:20 | 290 secs |
| 33 | AP-P2SC-SP [Repeat No.13] | 19-11-05 11:23 | 340 secs |
| 34 | AP-(P2SC-SP, P3LC-SP2CO2) | 19-11-05 12:30 | 340 secs |
| 35 | AP-(P1LC-SP, P2SC-SP) | 19-11-05 14:45 | 2880 secs |
| 36 | AP-(P1PC-SP, P1LC-SP) | 19-11-05 16:20 | 330 secs |
| 37 | AP-P1LC-CO [Repeat No.2] | 19-11-05 17:23 | 310 secs |
| 38 | AP-(P1PC-CO, P1LC-CO) | 19-11-06 08:58 | 310 secs |

## 63 Columns

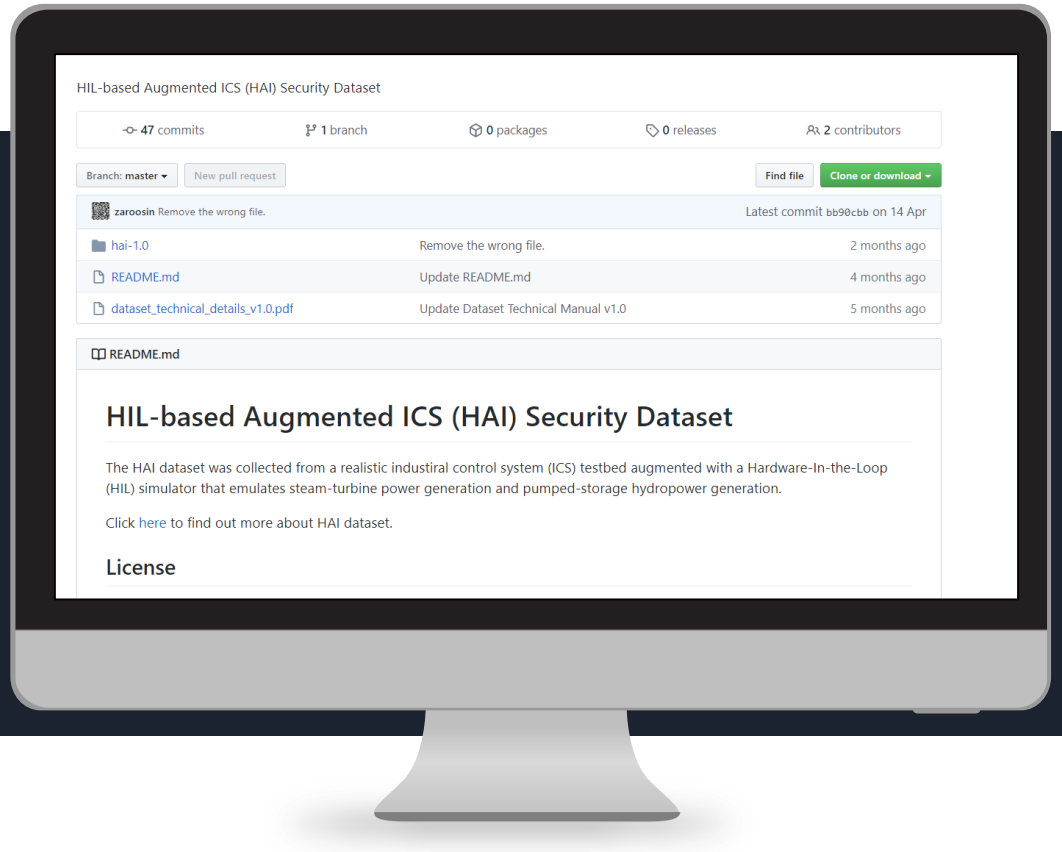
- **Column 01**: timestamp ‘yyyy-MM-dd hh:mm:ss’
- **Column 02 ~ 59**:
  - 58 data points continuously collected every second
- **Column 60**: attack label indicating for any attack
- **Column 61~63**: attack labels for each real system (boiler, turbine, water-treatment)

| time | P1.B2004 | P2.B2016 | ... | P4.HT_LD | attack | attack.P1 | ... | attack.P3 |
|---|---|---|---|---|---|---|---|---|
| 20190926 13:00:00+09:00 | 0.09830 | 1.07370 | ... | 0 | 0 | 0 | ... | 0 |
| 20190926 13:00:01+09:00 | 0.09830 | 1.07410 | ... | 0 | 1 | 0 | ... | 1 |
| 20190926 13:00:02+09:00 | 0.09830 | 1.07380 | ... | 0 | 1 | 0 | ... | 1 |
| 20190926 13:00:03+09:00 | 0.09830 | 1.07360 | ... | 0 | 1 | 1 | ... | 1 |
| 20190926 13:00:04+09:00 | 0.09830 | 1.07430 | ... | 0 | 1 | 1 | ... | 1 |
| #column01 | #column02 | #column3 | ... | #column59 | #column60 | #column61 | ... | #column63 |

# HAI 1.0 Security Dataset

**GitHub** https://github.com/icsdataset

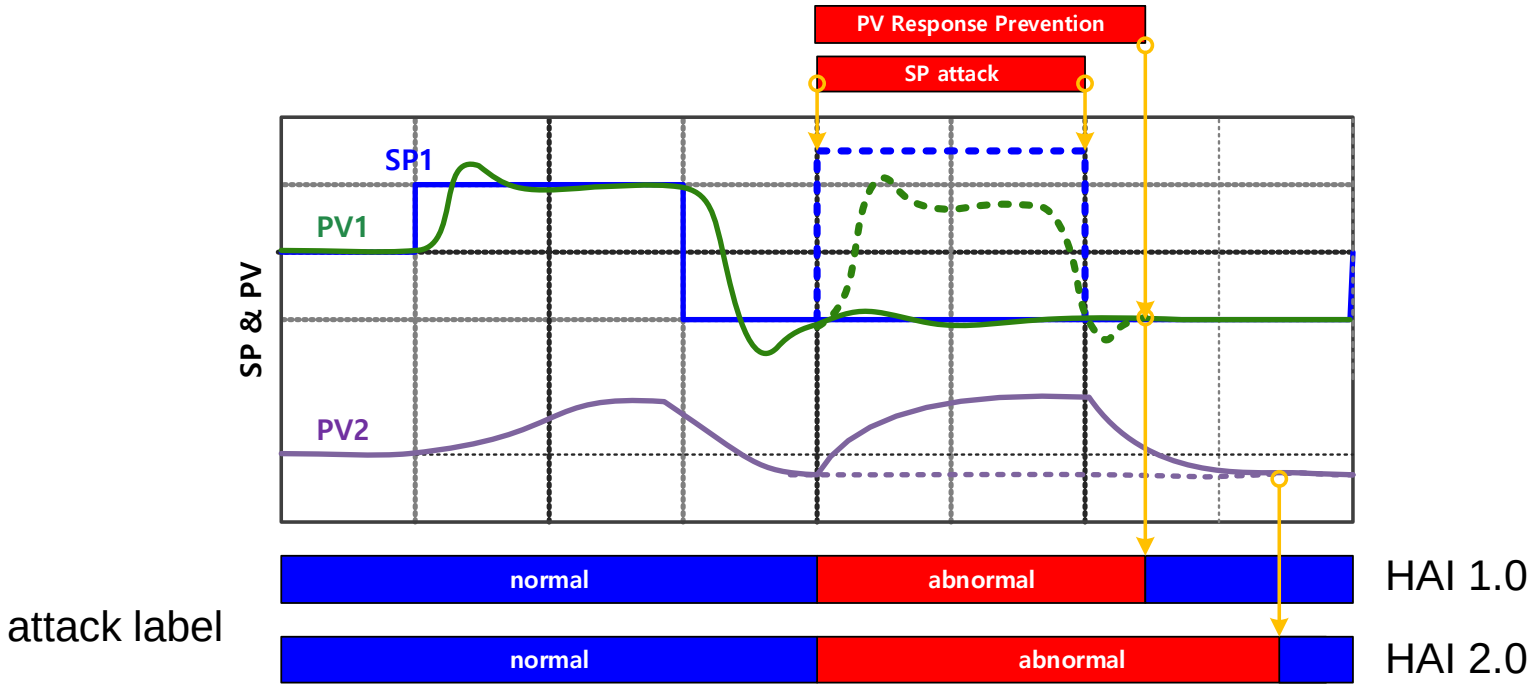
**Kaggle** https://kaggle.com/icsdataset



# Future Works

## Improving Data Labeling Method

- Including all transient sections according to attacks
  - A transient state identification(TSID) for the correlated PV values



# HAICon 2020

## Anomaly Detection Contest with HAI 2.0 Dataset

- ☑ Aug. 17 ~ Sep. 29
- ☑ ₩20,000,000 ($16,000) prize money
- ☑ https://dacon.io

**Please note that foreign participants must team up with at least one Korean**

